

AI-Powered Travel Package Recommendation and Booking Automation

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Problem:

Travel agencies receive a large number of inquiries every day. Customers ask for package details, trip costs, itinerary information, visa requirements, and booking support. Agents must manually calculate prices, gather data from multiple sheets, and send confirmation emails. This process is slow, prone to errors, and requires constant human availability. Customers also expect instant responses on the website, which traditional workflows cannot provide.

Solution:

To solve this, an automated AI-powered travel assistant workflow was created in n8n. It connects with the website frontend and handles the entire customer journey—from recommendations to booking confirmation. The system uses AI, RAG-based vector search, Google Sheets, and Gmail automation to provide accurate, real-time travel support.

Workflow Explanation:

The workflow is divided into two main parts: the Chat Workflow and the Knowledge Loader Workflow.

1. Chat Workflow (Customer Interaction)

This workflow activates whenever a customer sends a message from the website.

Nodes:

- Chat Trigger: Receives messages from the frontend chat widget.
- Workflow Configuration: Stores agent email and Google Sheet IDs.
- Google Gemini Chat Model: The AI brain that understands and generates replies.
- Conversation Memory: Keeps the conversation history for natural replies.
- Travel Knowledge Vector Store: Searches travel data (itineraries, visa rules, hotels).
- Pricing Data Tool: Fetches real prices from Google Sheets.
- Cost Calculator Tool: Calculates full trip cost including flights, hotels, taxes, and service fees.
- Booking Storage Tool: Saves confirmed bookings into Google Sheets.
- Customer Email Tool: Sends booking confirmation to the customer.
- Agent Notification Tool: Sends booking alert email to the travel agent.

This workflow handles:

- Travel package suggestions
- Pricing calculations
- Itinerary explanations
- Destination comparisons
- Multi-city recommendations
- Collecting customer booking details
- Confirming bookings by email

2. Knowledge Loader Workflow (AI Training System)

This workflow is used to load and update travel knowledge for the AI.

Nodes:

- Manual Trigger: Starts the data-loading workflow manually.
- Read Travel Data from Google Sheets: Reads all destination and package details.
- Format Travel Data: Converts each row into detailed text.
- Document Loader: Converts text into AI-readable documents.
- Text Splitter: Breaks large text into small chunks.
- OpenAI Embeddings: Turns text into vectors.
- Insert into Vector Store: Saves processed data into the AI's memory.

This workflow ensures that the AI knows:

- Destination overviews
- Day-by-day itineraries
- Visa requirements
- Hotel details
- Activities and attractions
- Best seasons and weather

Because of this separation, the frontend works smoothly. The chat workflow responds instantly without large processing delays.

Final Result:

With this automated system, customers receive instant answers, accurate pricing, and smooth booking support directly on the website. Travel agents get immediate notifications, and all bookings are safely stored. This workflow reduces workload, speeds up operations, and scales customer support without hiring extra staff.